Pathways, Carbon Budgets and Warming Outlooks

> Part of the Functional Series. The previous documents compared the platform's *observations* with benchmarks. This one compares its *scenarios* with the IPCC's emission pathways, sets out the reference carbon budgets (where the sources disagree), and compares the platform's implied warming with published outlooks. What these comparisons imply for the Paris targets is the subject of the Paris Agreement section (in development).

1. Reference pathways

The IPCC's AR6 Synthesis Report (Table SPM.1) gives the emission reductions from 2019 levels (median, with the 5th–95th percentile range) in modelled pathways that meet each goal.

Pathway	Gas	2030	2035	2040	2050
1.5 °C (>50%), no or limited overshoot	GHG	43 [34–60]	60 [49–77]	69 [58–90]	84 [73–98]
	CO_2	48 [36–69]	65 [50–96]	80 [61–109]	99 [79–119]
2 °C (>67%)	GHG	21 [1–42]	35 [22–55]	46 [34–63]	64 [53–77]
	CO_2	22 [1–44]	37 [21–59]	51 [36–70]	73 [55–90]

(Percent reduction from 2019.) UNEP's Emissions Gap Report 2025 uses its own pathway set and gives a slightly different benchmark for 2035: reductions of **35% (range 28–43) for 2 °C and 55% (range 50–63) for 1.5 °C**, in total GHG. The two sets are not interchangeable; the IPCC figures are used below.

2. The platform's scenarios against the CO_2 pathways

The platform's scenarios cover CO_2 only, so they are compared with the **CO_2** rows. Two scope points apply:

- IPCC CO_2 figures are **net of land use**; the platform's scenarios model fossil and cement CO_2 and **hold land use flat**. Both variants are shown; the "including flat land use" row is closer to the IPCC's definition, but the IPCC pathways also reduce land-use emissions.
- The platform's global figure is its 40 covered countries scaled up by holding the rest of the world at its 2024 share (10.7%).

Change in CO_2 from 2019 (2019 fossil CO_2 is 37.1 $GtCO_2$ and total including land use 42.0):

	2030	2035	2040
Observed 2024 against 2019	+4.1% fossil, +2.9% incl. land use		
Platform BAU, fossil	+10.6%	+15.9%	+21.0%
BAU, incl. flat land use	+8.8%	+13.5%	+18.0%
Platform Moderate (–2%/yr), fossil	–0.1%	–5.3%	–10.6%

	2030	2035	2040
Moderate, incl. flat land use	–0.6%	–5.2%	–9.9%
Platform Aggressive (–5%/yr), fossil	–14.4%	–30.6%	–43.9%
Aggressive, incl. flat land use	–13.3%	–27.6%	–39.4%
IPCC 2 °C (>67%), CO_2	*–22%*	*–37%*	*–51%*
IPCC 1.5 °C (>50%), CO_2	*–48%*	*–65%*	*–80%*

How to read this:

- **Observed emissions rose between 2019 and 2024**, while both IPCC pathways need falls. The pathways assume immediate action, which the IPCC defines as starting between 2020 and 2025.
- **Even the Aggressive scenario sits above the 2 °C median** in every year shown (for example –27.6% against –37% in 2035, and –39.4% against –51% in 2040), and far above the 1.5 °C median. This is a statement about the platform's *illustrative* scenario, not about what is achievable.
- **Moderate is roughly flat against 2019 in 2030** because its 2% a year is taken off a business-as-usual path that is still rising (+10.6% by 2030). A constant percentage cut compounds from the starting point of a growing pathway.
- **The IPCC range is wide.** For the 2 °C pathway in 2040 it runs from –36% to –70%; the platform's Aggressive scenario (–39% to –44%) is inside the 2 °C range but near its lower end.

3. Carbon budgets: where the references disagree

The remaining carbon budget is the cumulative CO_2 that can still be emitted for a given probability of holding warming to a level. The reference values differ because they are assessed at different dates, with different warming estimates.

Source	Start of budget	1.5 °C (50%)	1.7 °C (50%)	2 °C (50%)
IPCC AR6 (Synthesis Report B.5.2)	Beginning of 2020	500 $GtCO_2$		1,150 $GtCO_2$ (67%)
Forster et al. 2026 (Table 8)	Beginning of 2026	130	500	1,050
Global Carbon Budget 2025	Beginning of 2026	170	525	1,055

Reading across:

- **Forster et al. and the Global Carbon Budget agree on the larger budgets** (1.7 °C: 500 against 525; 2 °C: 1,050 against 1,055) and **differ by 40 $GtCO_2$ on 1.5 °C** (130 against 170). Both are small compared with the 2020 starting figure.
- **The IPCC's 500 from 2020 is not directly comparable** with the 2026 values by subtraction. Forster et al. put the total reduction since AR6 at about 370 $GtCO_2$, of which only part is emissions since 2020; the rest reflects revised warming estimates.
- **At 2025 emissions the 1.5 °C budget is nearly gone:** about 3 years at 42 $GtCO_2$ a year (Forster et al.) or about 4 years (Global Carbon Budget).

- **Using up the budget is not the same as reaching 1.5 °C.** Forster et al. state this explicitly. The values also cover only the uncertainty in the CO_2–warming relationship and assume a further 0.10 °C of non-CO_2 warming, which the platform does not model.

Any further figure for the 1.5 °C budget quoted from a secondary source should be checked against these primary values; at least one popular summary of the 2026 update quoted the 67%-likelihood value (80) as if it were the central estimate.

4. Warming outlooks to 2100

The platform does not project temperature beyond 2040, so these cannot be compared directly with its output. They show where the platform's short-horizon result sits in the wider picture.

Source	Scenario	Warming by 2100
IPCC AR6 WGIII	Policies implemented by end of 2020	3.2 [2.2–3.5] °C
IPCC AR6 Synthesis Report	NDCs announced before COP26, no further ambition	2.8 [2.1–3.4] °C
UNEP EGR 2025	Current policies	"up to" 2.8 °C
UNEP EGR 2025	Full implementation of all NDCs	2.3–2.5 °C
Climate Action Tracker (13 Nov 2025)	Current policies	2.6 °C
Climate Action Tracker	2030 and 2035 targets (NDCs)	2.6 °C
Climate Action Tracker	Pledges and targets incl. net zero	2.2 °C
Climate Action Tracker	Optimistic: all announced targets fully implemented	1.9 °C

The **two "2.8 °C" figures are different scenarios**: the IPCC's is for NDCs announced before COP26 (2021); UNEP's is for policies in place in 2025. Estimates range from about 1.9 °C to 3.2 °C depending on what is assumed to be implemented, and UNEP notes that methodological updates accounted for 0.1 °C of its own improvement from last year's 2.6–2.8 °C.

5. The platform's implied warming against published rates

	Platform (BAU, to 2040)	Forster et al. 2026
Level in 2025	1.34 °C	1.37 °C (human-induced)
Rate	about 0.23 °C a decade	0.27 °C a decade (2016–2025)
Year the level reaches 1.5 °C	2033 (BAU), 2033 (Moderate), 2034 (Aggressive)	around 2030 (at current emissions)

The platform's rate is lower and its crossing about three years later than Forster et al.'s. For context, the IPCC reports that the midpoint of the first 20-year average to reach 1.5 °C lies in the first half of the 2030s in all but its highest-emissions scenario, and gives 2030–2035 as the median five-year interval across the Working Group III pathway categories; the platform's 2033–2034 falls inside that window. Note that the IPCC's definition is a 20-year average, while the platform's "level" is a five-year-anchored annual value, so the match is indicative. This is the direction expected of a simplified model that uses only a CO_2 slope: it omits the additional warming from non-CO_2

gases and from falling aerosol cooling that the published rates include. The platform's figures should therefore be read as likely understating near-term warming, not as a central estimate.

6. What this comparison shows

- **The scenarios are illustrative bounding cases and fall short of Paris-aligned pathways.** Even the Aggressive scenario is above the 2 °C median CO_2 pathway.
- **Observed emissions have so far risen against the 2019 starting point** of the pathways.
- **Carbon budgets for 1.5 °C are nearly exhausted** under both current assessments, which differ by 40 $GtCO_2$.
- **Published outlooks span roughly 1.9–3.2 °C by 2100**, and the platform's short-horizon warming sits slightly below published rates.

See also

- The Reference Landscape
- Emissions Compared with Benchmarks
- Temperature and Concentration Compared with Benchmarks
- Scenarios and Temperature Mapping